

# Kartik Namdev

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## Education

**Vellore Institute of Technology, Vellore**

**2023 – 2027**

*B.Tech. in Computer Science & Engineering (IoT Specialization) — CGPA: 9.13 / 10*

## Experience

**Samsung R&D Institute India – Bangalore**

**May 2026 – Present**

*Research and Development Intern*

**Project: Autonomous Interaction Management Framework for Dynamic Site Handling**

- Engineered a hybrid browser automation framework using Playwright, FastAPI, and LLMs with DOM-first execution, reducing visual overhead by 40% and cutting execution latency by 25%.
- Architected a semantic DOM analysis and RAG pipeline to resolve unanchored web elements, boosting autonomous workflow execution reliability to 95%+ across dynamic target applications.
- Optimized multi-step reasoning via semantic Top-K candidate ranking of interactive elements, filtering DOM nodes against user intent to lower LLM token consumption by 35%.

**Tech Stack:** Python, Playwright, FastAPI, LLMs, RAG, Prompt Engineering, Browser Automation

## Projects

**CareerPilot AI – AI-Powered Interview Preparation Platform**

 [GitHub](#)

*React.js, Node.js, Express.js, MongoDB, JWT, Google Gemini, Puppeteer, Multer*

- Engineered an end-to-end AI interview preparation platform leveraging Google Gemini with Zod schema validation to analyze resumes against job descriptions, generating personalized match scores, skill-gap reports, and role-specific interview questions in under 2.5 seconds.
- Architected secure REST APIs featuring JWT token blacklisting, Multer-based document ingestion, and Puppeteer PDF rendering, backed by indexed MongoDB collections enabling sub-80 ms historical report retrieval.

**StarTube – Interactive Media Sharing Platform**

 [GitHub](#)

*Node.js, Express.js, MongoDB, JWT, Cloudinary, Multer*

- Developed a modular RESTful backend supporting secure authentication, multi-format media uploads, playlists, subscriptions, comments, and watch history through Cloudinary integration and Multer memory storage.
- Optimized database retrieval across subscription feeds and watch history using MongoDB aggregation pipelines and index tuning, reducing API query latency by 30%.

## Skills

- Programming Languages:** C++, Java, Python, JavaScript, SQL
- Backend:** Node.js, Express.js, FastAPI, RESTful APIs, JWT Authentication, Microservices
- Frontend:** React.js, Redux Toolkit, HTML5, CSS3, Responsive UI Design
- Databases:** MongoDB, MySQL
- DevOps & Cloud:** CI/CD Pipelines, Docker, Kubernetes, Jenkins, AWS, Linux
- Artificial Intelligence:** Large Language Models (LLMs), Prompt Engineering, RAG, LangChain
- Tools & Testing:** Git, GitHub, Postman, Playwright, Puppeteer, VS Code

## Certificates

- Oracle Cloud Infrastructure 2025 Certified DevOps Professional – Oracle 
- Generative AI using IBM Watsonx – IBM 
- Python Foundation Certification – Infosys 
- Database Management System – Infosys 